

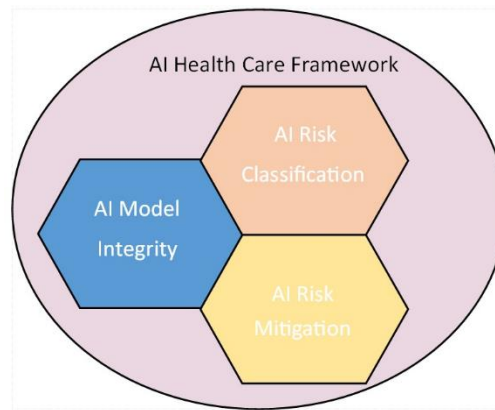
### AI Ethics Framework in Healthcare Industry

A transparent and unified AI Ethical Framework is an essential component behind any successful Enterprise or Industry. Modern Artificial Intelligence (AI) Systems have been extensively engineered to design algorithms to independently classify and cluster data (Carter, Rogers and Win). These algorithms have the capacity to develop, by exposure to data, the ability to recognise patterns in that data. Consequently, these patterns can assist in the extensive training of AI models and thus, help to accurately predict the possibility of a patient suffering from a particular disease. For instance, the use of an AI model would prove highly efficient in situations where accurately analyzing ultrasound images, mammograms, and MRI data (Kaulitzki) contribute in reducing the frequency of false positives during medical diagnostic process.

AI models, although proven to produce highly accurate outcomes, are not exempt from ethical risks, such as Data Bias, Data breaches and misuse among others. Failing to address AI ethics could attract a series of ethical, not to mention, legal implications for the company or government body involved. For example, DeepMind, a Google-owned Artificial Intelligence Enterprise, is currently facing a class-action lawsuit over its controversial use of NHS patients' health data back in 2015. According to source, 1.6 million of NHS patient medical records were secretly given to Deepmind for AI Health research purposes (Daws).

The health care Industry, for instance, has been a mayor ethical focus point for Systematic ethical risk Mitigation since the early 1970's (Blackman). An Industry whose main purpose derives from its direct contribution to the physical and psychological well being of the human individual, must rely on strong ethical, moral, and legal parameters. Medical ethicists, lawyers, and health care practitioners have invested resources in establishing AI frameworks to act as guidelines for key ethical concerns. In this case, a complete AI Healthcare framework has been designed to incorporate all these important parameters and focuses on three main components: AI Model Integrity, AI Risk Classification, and AI Risk Mitigation.

The AI Model Integrity captures a strong set of ethical principles to train AI Models, and ultimately contribute to a transparent and just system. For instance, The Montreal Declaration (Montreal Declaration) for Responsible AI Development has developed an AI Ethical Framework that identifies ethic principles from which an AI Model should be based on. The abstract principles showcase the main values that promote the fundamental interests of individuals and groups. The Well-being Principle prioritizes the notion that AI must always assist in the improvement of an Individual's living and health conditions. The protection of the physical and mental self is fundamental for an AI Model if it also respects their clear and explicit consent and understanding and also prevents harm to another being. In addition, AI should not contribute to the deterioration of an individual's physical, mental, and otherwise psychological well-being. In addition, another important principle is Respect for Autonomy. AI Model must be designed and tailored to condition respect for individual autonomy, allowing each individual to exercise their fundamental rights with regards to their health and personal data privacy, including clear and informed consent decisions towards treatment and diagnostic procedures. AI must not impose, manipulate, or otherwise coerce any decision or lifestyle, whether directly or indirectly. Under no circumstance, should AI be developed to provide or spread untrustworthy information, manipulative language, or incoherent communication (Impenetrable language) that poses as a barrier for an individual's capacity to understand. In an Industry where frequent decisions regarding an individual's quality of life or reputation may be affected; it is important to incorporate a Responsibility Principle. This clause insinuates that all human actions and decisions that proceed from AI outcomes and recommendations are subject to full human accountability. Under any circumstance in which a decision must be made regarding an individual's quality of life, health, or reputation, under considerable exemptions, the final decision must always stem from an informed and free human decision. However, in a rare and given situation where harm has been inflicted by an AI, whose system has been deliberately proven to be used as intended, no blame should be placed on the individuals involved in its development and/or use. The last of the studied principles, the Solidarity Principle, emerges from a unified ethical model. The design and use of AI system should be intended to contribute towards the creation of a data transparent and equitable society. It is important to implement anti-discriminatory values and policies into the algorithm training to allow for unbiased and effective data privacy protection policies. This principle supports equitable and unbiased AI recommendations and outcomes that otherwise affect an individual's personal self, quality of life and patient care. By identifying and implementing these abstract ethical principles into the algorithms involved in AI System model training, it can act as a guide for classifying toolkit strategies for key ethical concerns.



Although, it is crucial to define a solid AI Model Integrity, all AI Models are subject to ethical risks. Classifying ethical risk zones is an important step toward operationalizing data and AI ethics, by providing a transparent and structured approach that allows for AI risk mitigation. According to the Ethical 8 Risk Zone Framework by IFTF (Ethical OS Toolkit), a total of eight main risk zones are identified to help cluster and classify all key ethical concerns resulting from AI Model outcomes. The Risk areas found most relevant with regards to its impact in the health Industry are the following: Risk zone 1: Truth, disinformation and propaganda, Risk Zone 2: Implicit trust and User Understanding, Risk Zone 3: Data control and monetization, and Risk Zone 4: Machine Ethics and Algorithmic Biases.

With a classified set of clearly defined risks, it becomes easier to create an effective AI Risk Mitigation Plan to counteract those key concerns. The following AI Risk Mitigation component of the framework is based on the AI Algorithmic Accountability Policy Book, an open-source bias audit toolkit developed to audit machine learning models for discrimination and Data Bias (AI Now Institute).

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#### AI Healthcare Ethics Framework

In January 2021, there are more than 3.8 million women with a history of breast cancer in the United States (Kaulitzki). In addition, the prediction for 2021 in Canada is 27,700 women will be diagnosed with breast cancer which represents 25% of all new cancer cases in women in 2021. In Canada, on average 15 women die from breast cancer (Canada Cancer Society). Predicting it on time could save many lives. Nowadays, there are AI algorithms that helps us to predict it on time and reduce false positives, but how can we ensure that these algorithms comply with AI ethics? This case of study will be applied in “AI Health Care Framework” in order to verified whether the algorithm complies with the AI ethics of the designed framework. The framework is divided in 3 phases where the first phase is about principles, the AI algorithm must be guaranteed the improvement on how the person lives after the breast cancer detection, giving them an opportunity to get a non-invasive treatment which correspond to the principle of “well-being”. Then, the final decision to ensure that the cancer is a fact that is happening, must be decided for a doctor and in this way the principle of “accountability” would be fulfilled. Furthermore, the algorithm must be trained with different datasets in order to avoid gender, skin or aged bias, so the principle of “solidarity” will be satisfied. Finally, the last principle “Respect for autonomy” must be comply when the patient is asked in a meaningful way in order to use this new proceed or AI algorithm. However, there are many risks associate of using AI in health care. One of the risks would loss of trust, clinic negligence, overdiagnosis, errors on predicting breast cancer which are classified in Zone 1. Another examples of risk of misunderstanding, entirely illusory consent would be classified in Zone 2. For instance, the use of old data, prescribe wrong medication, gaps in cybersecurity and data privacy are risk which belongs to Zone 3. Finally, risk associated to gender, color skin bias and so on are classified in Zone 4. In order to mitigate these risks, the framework propose the following solutions such us implicit trust and user understanding, data collection transparency and consent, keep updating Cybersecurity and Confidentiality policies over the time, mandate validation study, prescription drug monitoring databases and Health Surveillance Data.

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